

24-502-0103: Design and Analysis of Algorithms

Core/Elective: **Core**

Semester: **1**

Credits: **4**

Course Description

The course covers the foundational algorithms in depth. The course helps in understanding the working and complexity of the fundamental algorithms and to develop the ability to design algorithms to attack new problems.

Course Outcomes (CO)

After the completion of the course, the students will be able to:

CO	Course Outcome Statement	Cognitive Levels
CO1	Understand the basic concepts of design and analysis of fundamental algorithms.	Understand
CO2	Develop the ability to design algorithms to attack new problems.	Apply
CO3	Prove the correctness of algorithms.	Analyze
CO4	Develop the ability to analyze the complexity of algorithms.	Analyze
CO5	Understand Complexity classes, concepts of P and NP problems	Understand

Mapping with Program Outcomes

	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	-	3	-	3	3
CO2	3	-	-	3	3	-	3	3
CO3	-	3	-	3	3	-	3	3
CO4	3	3	-	-	3	-	3	3
CO5	3	-	-	3	3	-	3	3

Course Content

1. Introduction to design and analysis of algorithms, models of computation, correctness proofs, insertion sort, computational complexity, Master theorem , proof of Master theorem, merge sort, heaps, heap sort, binary search, binary search trees.
2. Graph algorithms, BFS and DFS, Dijkstra's algorithm, proof of correctness of Dijkstra's algorithm, Complexity analysis of Dijkstra's algorithm , Negative weight edges and cycles , Bellman-Ford algorithm, proof of correctness and complexity of Bellman-Ford, All pairs shortest paths, Floyd- Warshall algorithm, proof of correctness and complexity, Minimum Spanning Trees , Prim's algorithm, Cut property, Kruskal's algorithm, proof of correctness and complexity analysis of Kruskal's Algorithm, Maximum-Flow networks, Ford-Fulkerson method, proof of correctness and complexity, Edmonds-Karp algorithm
3. Probability review, Experiments, outcomes, events, Random variables, Expectation, Linearity of Expectation,

Indicator Random Variables, Hiring Problem, Quicksort , Best case and Worst case complexity, Randomized Quicksort , Average case complexity , Hashing, Chaining, Open Addressing, Universal Hashing, Perfect Hashing , Analysis of hashing operations

4. Dynamic Programming , Rod-cutting problem, Recursive formulation, Bottom-up reformulation of recursive algorithms, Optimal Substructure Property, Matrix chain multiplication, Complexity of dynamic programming algorithms, Sequence Alignment , Longest common subsequence, Greedy algorithms, Optimal substructure and greedy-choice properties , 0-1 and fractional Knapsack problems, Huffman coding
5. P vs NP, NP Hardness, Reductions, Travelling Salesman Problem, NP-Completeness, SAT, 2- SAT and 3- SAT, Vertex Cover

References

1. Michale T Goodrich and Roberto Tamassia, Algorithm Design and Applications, Wiley, 2014
2. Thomas H. Cormen et al, Introduction to Algorithms, MIT Press; 4th edition 2022.
3. Jon Kleinberg, Eva Tardos, Algorithm Design, Pearson; 1st edition August 2013.
4. Robert Sedgewick, Kevin Wayne, Algorithms, Addison Wesley; 4th edition 2011.
5. Steven S. Skiena, The Algorithm Design Manual, Springer; 3rd ed. October 2020